



Deep Learning for Short-Horizon Market Volatility Prediction

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Evaluating sequence models for SPY volatility forecasting under realistic temporal constraints

01. Background & Motivation

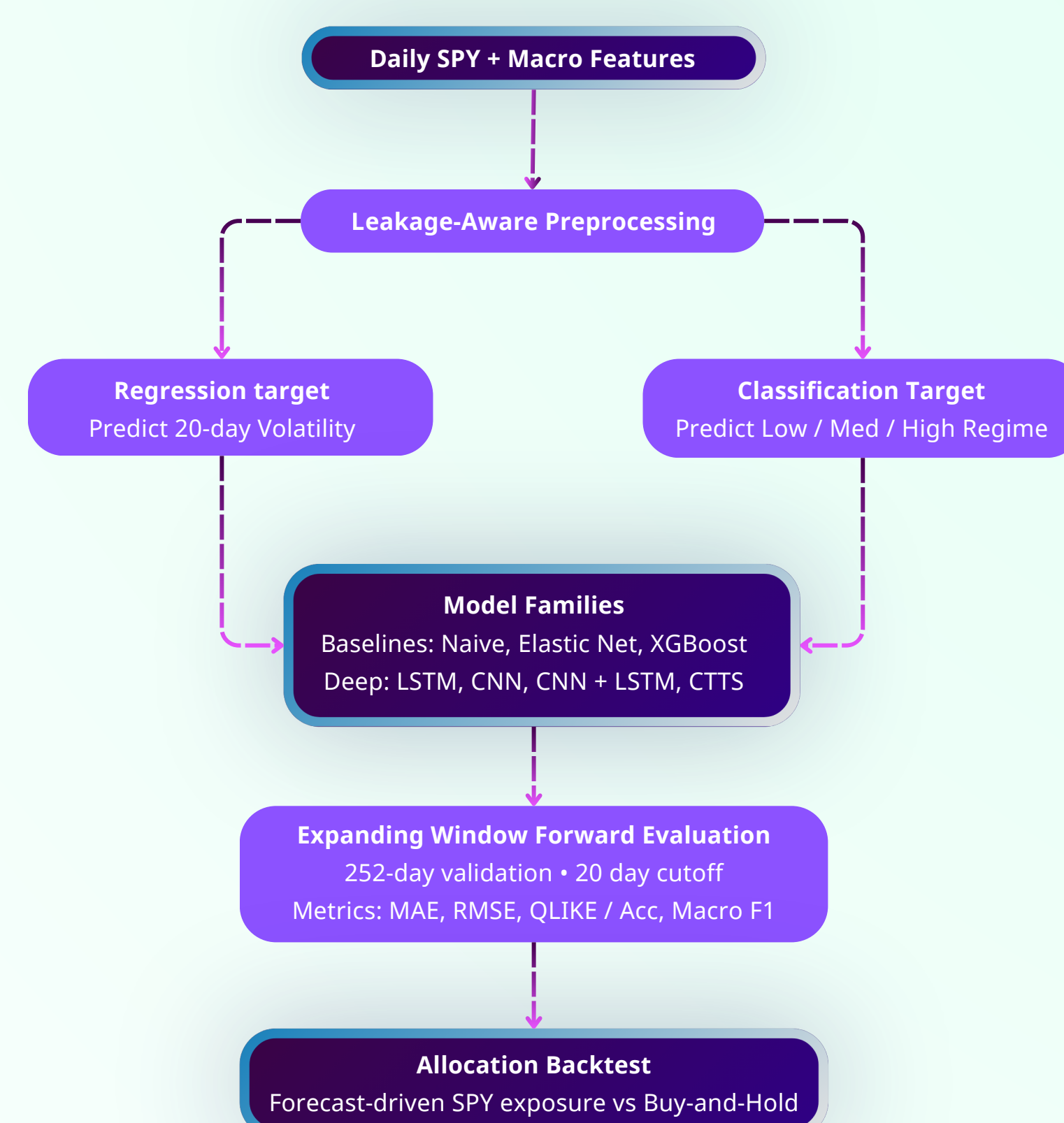
- Stock return prediction is noisy, but volatility is more useful for risk-aware allocation.
- We study whether short-horizon volatility forecasts for SPY can improve adaptive portfolio decisions.
- We compare two forecasting setups:
 - Regression:** predict future 20-day realized volatility
 - Classification:** predict low, medium or high future volatility regims
- Research Question:** Do deep sequence models improve volatility forecasting under realistic forward evaluation?

02. Data

- Daily SPY-centered dataset with **24 market features**
- Features include SPY prices, returns, volume, price ranges, VIX, CPI etc.
- Target is the **future 20-day realized volatility** of SPY
- For classification specifically, we classified the data to predict **low, medium or high** future volatility

03. Methodology

- Compared classical baselines against deep sequence models:
 - Baselines:** Naive persistence, Elastic Net, XGBoost
 - Deep models:** LSTM, CNN, CNN+LSTM, CTTS / CNN-Transformer
- User **expanding-window walk-forward evaluation** to mimic real forecasting conditions
- Deep models were tuned with **Optuna** and **early stopping**
- Regression metrics:** MAE, RMSE, QLIKE
- Classification metrics:** Accuracy, Balanced Accuracy, Macro F1
- Forecasts were also evaluated through downstream allocation **backtests**

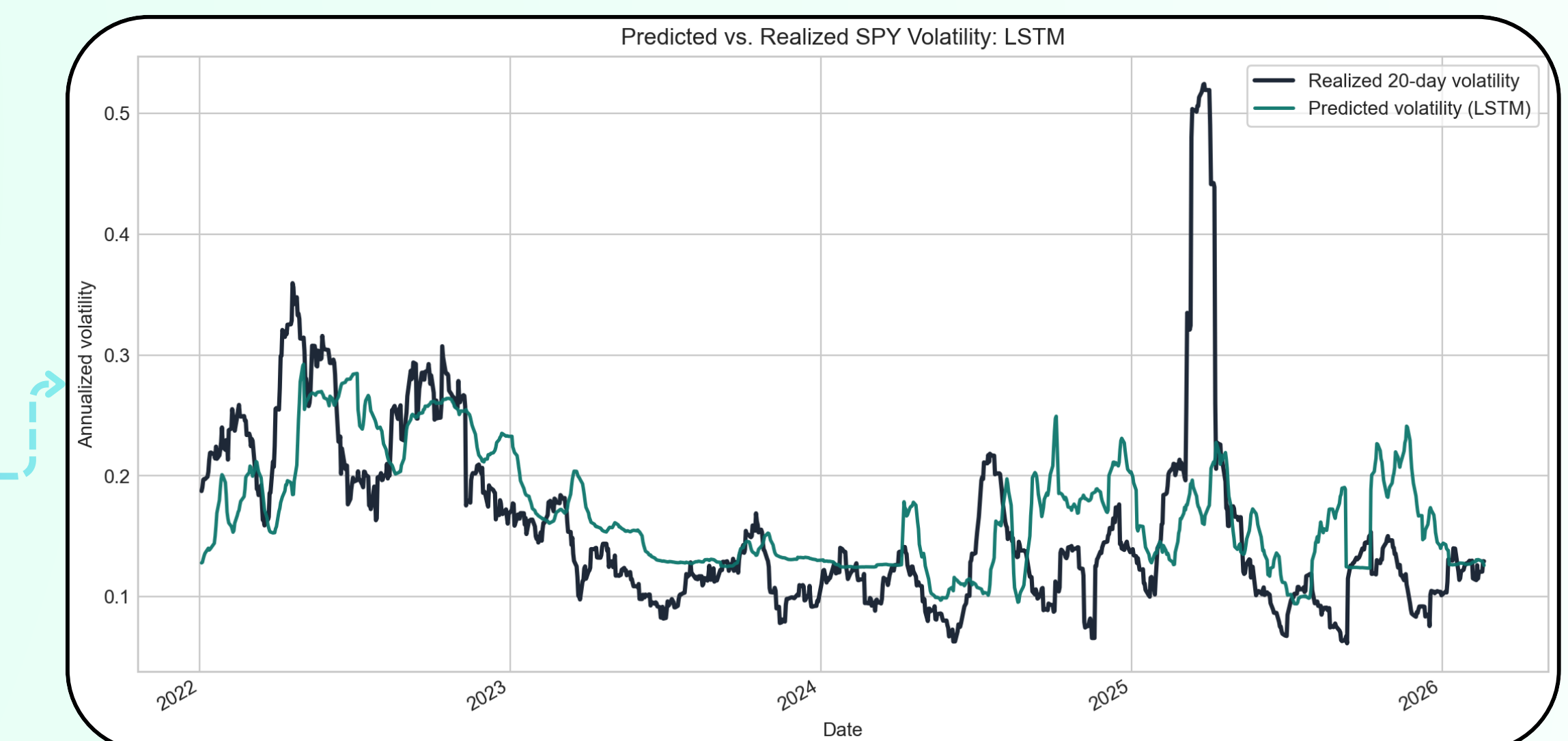
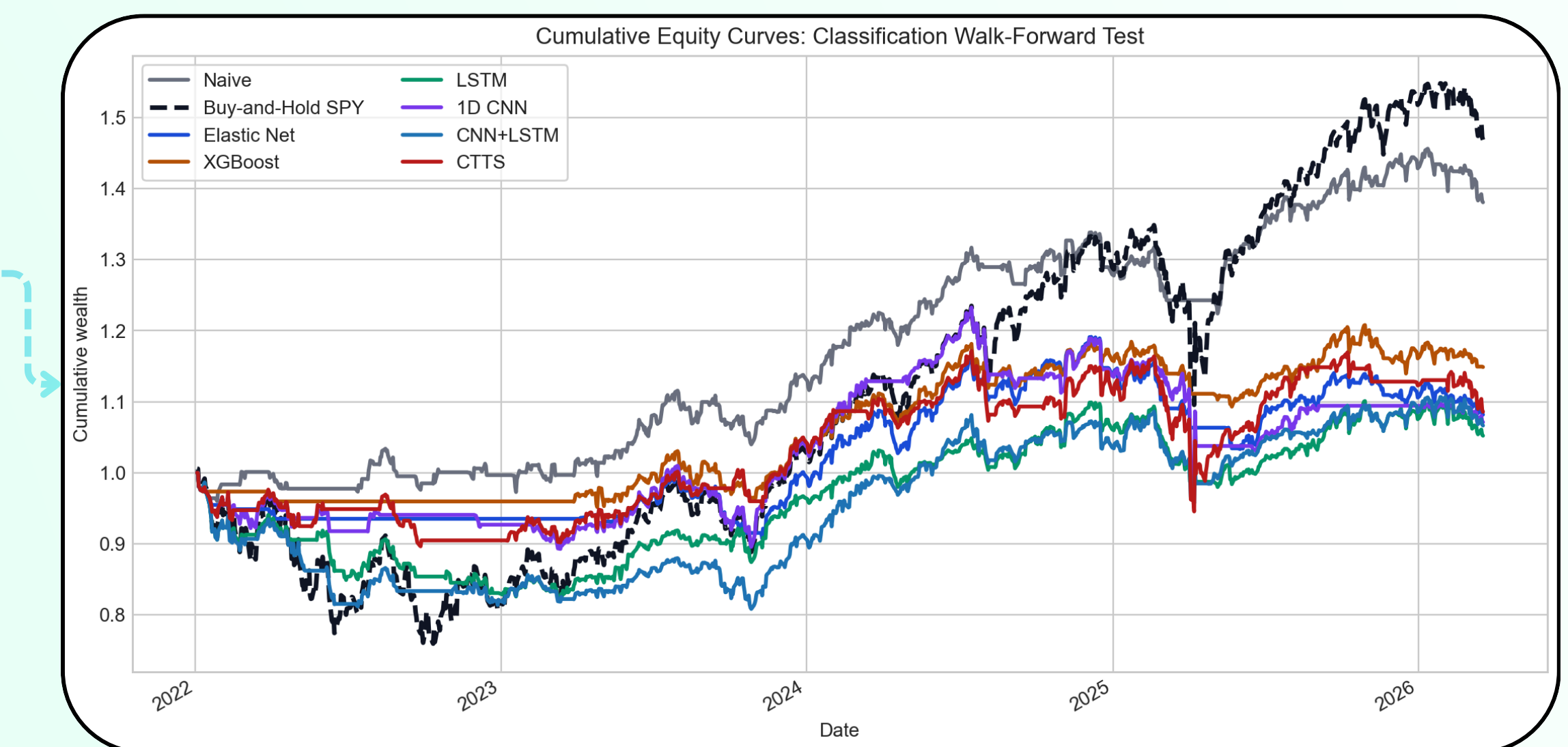


04. Results/Findings

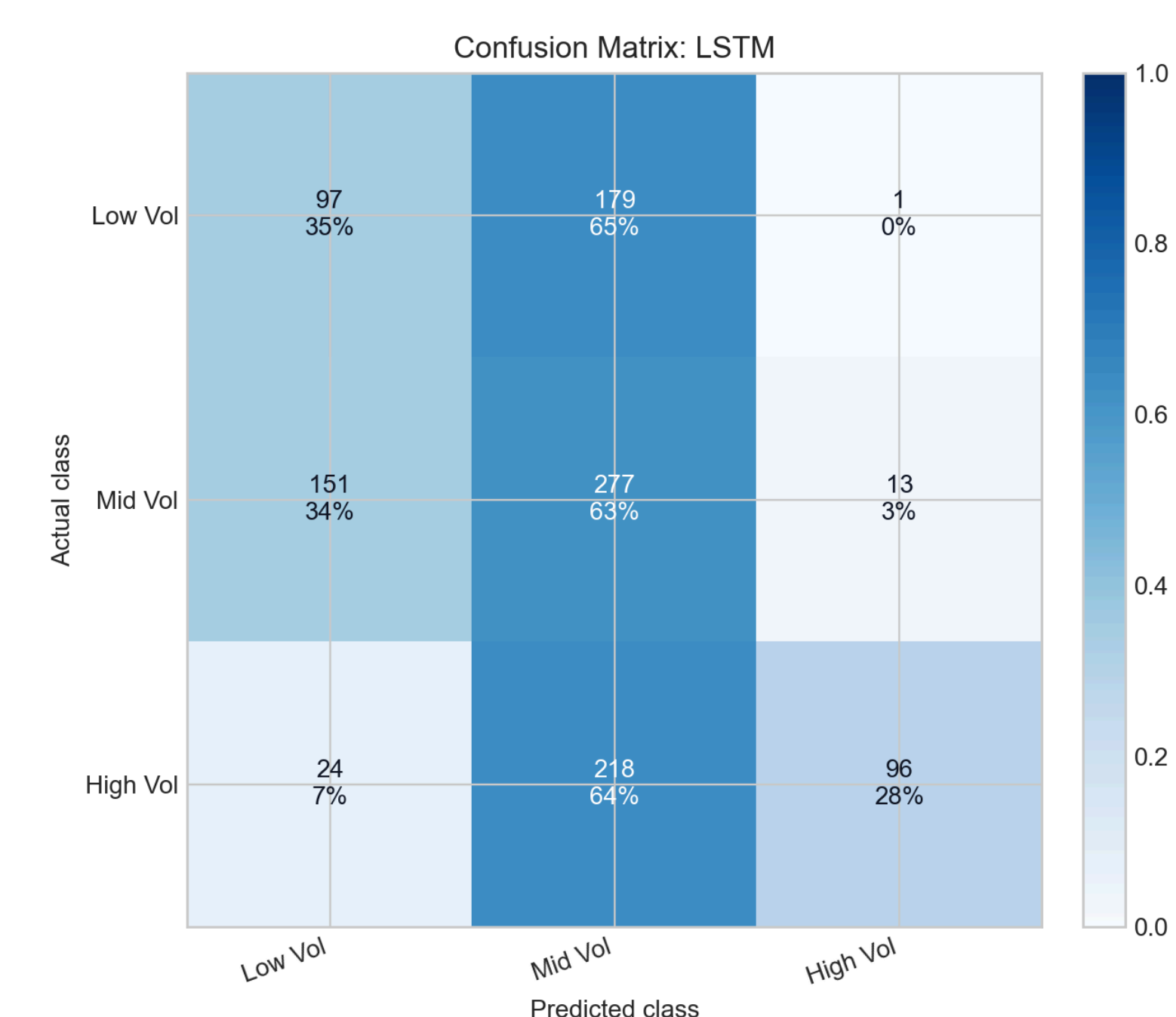
- Elastic Net achieved the best overall regression error**, with the lowest MAE and RMSE
- XGBoost achieved the best QLIKE, making it a strong volatility-specific benchmark
- In classification, **Elastic Net** achieved **best accuracy, balanced accuracy and macro f1**
- Among deep models, **LSTM** was the most competitive overall, but still trailed the typical models
- CNN, CNN+LSTM and CTTS showed weaker generalization, likely due to limited daily data and high noise
- Deep learning diagnostics suggest **signs of overfitting** and unstable **generalization**
- Allocation strategies captured some volatility structure but did **not outperform buy-and-hold SPY**

The top figure shows the trading strategy compared to buying and holding the SPY. Compared to baselines, our model does not have a substantial difference, indicating that the strategy might be the limiting factor.

The bottom figure shows predicted realized volatility compared to the SPY (specifically of the LSTM). Note that there is a strong correlation of trends, except for a few notable spikes (e.g Q1 2025)



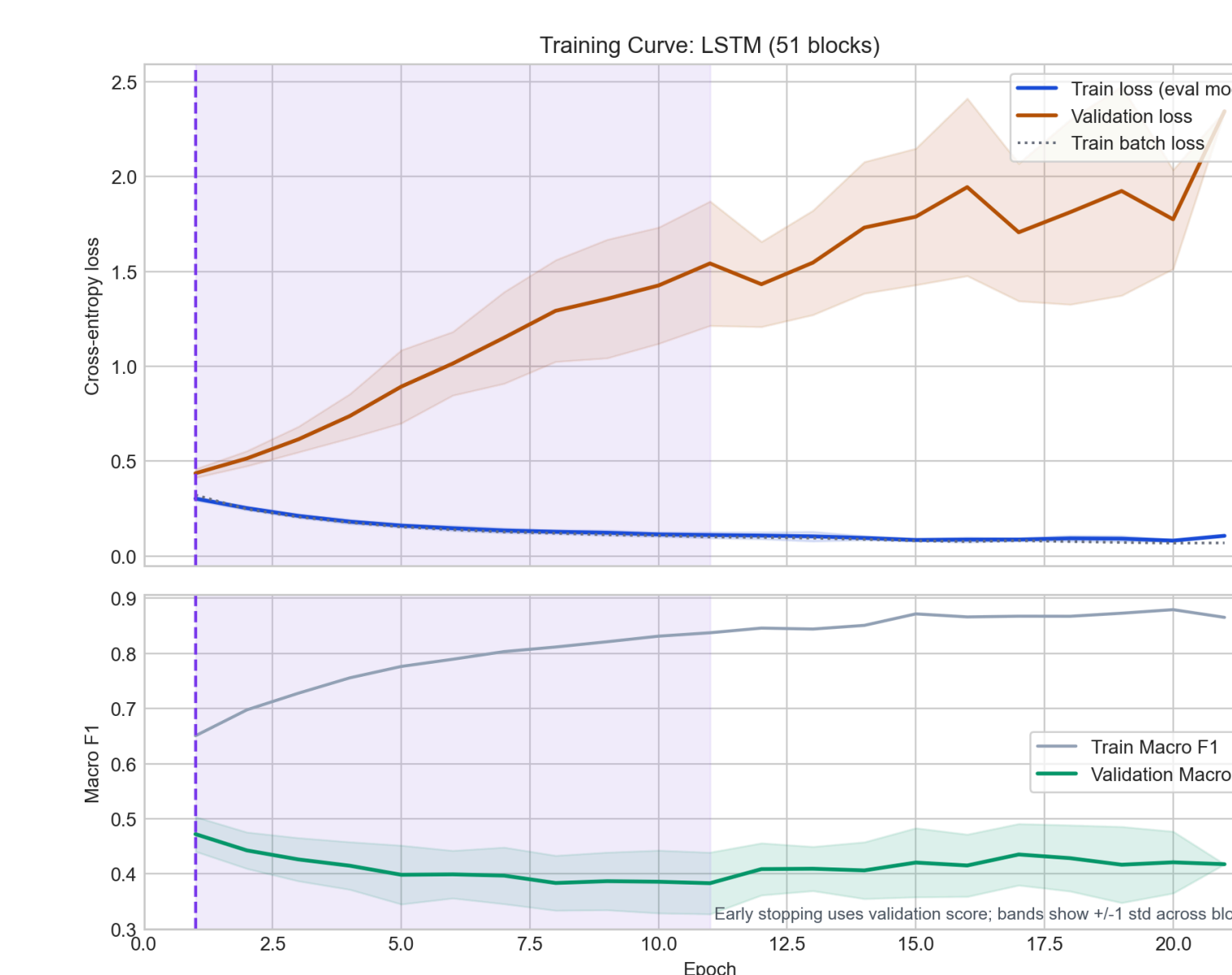
Model	Accuracy	Balanced Accuracy	Macro F1
Naive	0.5087	0.5311	0.5150
Elastic Net	0.6052	0.6293	0.6063
XGBoost	0.5743	0.5898	0.5746
LSTM	0.5154	0.5517	0.5241
CNN	0.4566	0.4789	0.4625
CTTS	0.3205	0.3335	0.3224



Figures showing the F1 scores, confusion matrices and training scores (specifically for our LSTM models).

Although Deep Models weren't as competitive as the standard models, the LSTM was the most optimal.

Additionally, note the increase in validation loss due to our lack of data and thus to mitigate that we introduced early-stopping.



05. Future Work

- Add **richer financial features**, including intraday data, options-implied volatility and other sector-level signals
- Incorporate **textual/event-based data** such as news, earnings headlines and central bank communication
- Test simpler and more regularized neural architectures
- Improve **trading evaluation** with transaction costs, alternative volatility targeting rules and regime specific allocation polices

06. References

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